



1.3 Financial applications of geometric sequences

Number and algebra · IB Diploma · Concept guide and original exam-style practice

Understand the concept

Compound interest is modelled by a geometric sequence: an amount P invested at nominal annual rate $r\%$, compounded k times per year for n years, grows to $FV = P(1 + r/(100k))^{(kn)}$. Depreciation uses the same structure with a rate less than 1 (value multiplied by $(1-r/100)$ each period). These are direct financial uses of $u_n = u_1 \cdot r^{(n-1)}$ type growth/decay.

Worked example: follow the reasoning

\$2000 is invested at 6% p.a. compounded monthly for 4 years. Find the final value.
 $FV = 2000(1 + 0.06/12)^{(12 \times 4)} = 2000(1.005)^{48} = 2000(1.2705) = \2541.03 (2 d.p.)

Quick check · multiple choice

A car worth \$30000 depreciates at 15% per year. What is its value after 2 years?

- A \$21675
- B \$25500
- C \$19200
- D \$22950

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 An investor deposits \$5000 into an account paying a nominal annual interest rate of 4.8%, compounded quarterly. [6]
- (a) Find the value of the investment after 5 years.
 - (b) Find the minimum number of whole years needed for the investment to exceed \$7000.

Final answer: _____



Answers and explanation

Question 1 [6]

(a) $FV = 5000(1 + 0.048/4)^{(4 \times 5)} = 5000(1.012)^{20} = 5000(1.2682) = \6341.15 .

(b) Need $5000(1.012)^{(4n)} > 7000 \rightarrow (1.012)^{(4n)} > 1.4 \rightarrow 4n \cdot \ln 1.012 > \ln 1.4 \rightarrow 4n > 28.24 \rightarrow n > 7.06$, so $n = 8$ years.

Quick check answer

D. $30000 \times 0.85^2 = 30000 \times 0.7225 = \21675 ... recheck: $0.85^2 = 0.7225$, $30000 \times 0.7225 = 21675$.

Correct answer is A.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Mathematics AA subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.